

AVENTRO



Building the Next Generation of Physical AI Through World Models

Epistemology-driven, zero-shot generalization for manufacturing, logistics, and smart cities.

Target Program: Startup Qatar

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Current Physical AI is Brittle, Expensive, and Unsafe.



Data-Hungry

Existing robotic vision systems require massive, labeled datasets to function, making deployment in novel environments slow and costly.



Lack of Causal Understanding

Traditional deep learning pattern-matches; it doesn't understand why things happen, leading to catastrophic failures in unpredictable physical spaces.



The Safety Gap

No built-in mechanism to say "I don't know," resulting in high false-positive rates or dangerous physical actions in industrial settings.



The Infrastructure Bottleneck

Most AI researchers lack production MLOps experience, so cutting-edge research rarely survives contact with real-world robotic middleware (ROS2).

THE SOLUTION

Software Robotics Powered by Epistemology-Driven World Models

Aventro bridges the gap between theoretical AI and production-grade robotics. We don't just build models — we build machines that reason about the physical world.



Epistemology-First Reasoning

Models that inherently reason about what they know, what they don't know, and how to acquire new knowledge efficiently.



Geometric Field Prediction

JEPA-inspired: predicts future states in representation space, not pixel space — capturing geometry, affordances, and causality.



Uncertainty-Aware Architectures

Proprietary Bayesian normalization (R-LayerNorm) ensures the system knows when it is uncertain, enabling safe shutdowns or human handovers.



Production-Ready ROS2 Stack

Built from day one to integrate with existing robotic infrastructure — not just run in a lab.

Why Aventro Wins

1

Zero-Shot Generalization

Deploy to new factories, warehouses, or infrastructure sites without expensive retraining or data labeling.

2

Provable Safety

Our SafePredict “shadow monitor” catches physical-AI failures before they reach robot actuation.

3

Resource Efficiency

State-of-the-art Physical AI doesn't require million-dollar GPU clusters — it requires superior math. Validated on free-tier T4/P100 GPUs.

4

End-to-End Delivery

Unlike pure research labs, our founding team has 10+ years of IT infrastructure and MLOps experience at scale.

How We Make Money

Aventro operates a B2B Enterprise SaaS & Custom R&D model.

PRIMARY

Enterprise Licensing

Annual licensing for our Aventro Edge SDK (perception, world-model prediction, safety monitor) deployed on client edge servers/robots.

EARLY STAGE

Custom R&D Pilots

Project-based fees to adapt our base architecture to specific client use cases — e.g., custom geometric quality control for a specific car part.



Pricing Strategy

Tiered licensing based on the number of edge nodes/robots and compute requirements, transitioning to consumption-based API pricing for cloud inference.



Cost Structure

Primarily R&D talent, cloud compute (optimized via our efficient architectures), and targeted B2B enterprise sales.

MARKET SIZE

Capitalizing on Qatar's Industrial AI & Logistics Transformation

\$18.5B

TAM — GLOBAL

Global Physical AI & Computer Vision in Manufacturing/Logistics market by 2028.

\$4.1B

SAM — GCC

GCC's rapid adoption of industrial automation — Saudi Vision 2030, Qatar National Vision 2030, UAE Operation 300bn.

\$380M

SOM — QATAR FOCUS

Addressable market in Qatar manufacturing, logistics, energy infrastructure, and smart-city systems over 5 years.

WHY QATAR

1

Massive post-World Cup smart infrastructure requiring intelligent monitoring and maintenance.

2

Hamad Port's ambition to become a global top-10 logistics hub — demanding warehouse automation and autonomous systems.

3

QatarEnergy's push for AI-driven predictive maintenance and remote inspection across LNG and upstream facilities.

4

Qatar Free Zones offer a frictionless launchpad for deep-tech companies targeting the wider GCC market.

The Aventro Moat

Where We Sit

Top-right: the only player combining deep theory with real production MLOps — not a lab demo, not a rules engine.



Provable geometric & epistemic theory + production MLOps at scale.

↑ PRODUCTION READINESS

Traditional CV <i>(Cognex, etc.)</i> High production readiness, low theoretical sophistication. Requires rigid rules and massive labeled data.	Aventro The only player combining provable geometric/epistemic theory (8 first-author papers) with deep production MLOps expertise (Kafka, Kubeflow, TensorRT, 24-GPU cluster mgmt).
Generic MLOps Platforms Infrastructure only — no proprietary physical AI models.	Big Tech Research <i>(DeepMind, Meta FAIR)</i> High theoretical sophistication, low production/ROS2 readiness. Focus on benchmarks, not edge deployment.

Proof of Execution Speed and Depth

1

Academic / Research

Authored 8 first-author preprints forming a cohesive “Physical & Geometric AI” methodology in under 10 weeks.

Validated entirely on free cloud GPUs — proving extreme capital efficiency.

Includes novel geometric theorems and Bayesian architectures.

2

Production (Prior Roles)

Shipped multimodal healthcare AI: F1 +19%, Latency -30%.

Fraud/IoT anomaly detection: AUC +17%, False Positives -22% ($p_{99} \leq 95\text{ms}$ at 500 RPS).

Built InsurTech PoC: fine-tuned SAM 3 for drone roof damage with custom mIoU-to-cost pipelines.

3

Open Source / Community

SafePredict white paper bridging LeRobot v0.6.0 with enterprise MLOps (Triton/Kafka).

Active Kaggle presence.

Open GitHub repositories.

Current Status & The Ask

FUNDS RAISED TO DATE

\$0 — 100% Bootstrapped / Founder Capital

CORPORATE STATUS

Aventro UK Co. (Incorporated)

THE TEAM

Two-person founding team — AI/Software (Mohsen) + Hardware/Electrical (Kaiz) — end-to-end physical AI delivery.

THE ASK

Startup Qatar Program Benefits

Grant funding, mentorship, ecosystem access, and support with Qatar entity establishment.

USE OF STARTUP QATAR PROGRAM BENEFITS

45%

Talent

Hire 1-2 Senior Robotics/AI Engineers locally in Doha, leveraging Qatar's deep-tech talent pool from HBKU, QCRI, and Qatar University.

30%

Hardware & Compute

NVIDIA Jetsons, sensors for pilots with Qatari industrial partners + cloud compute for training.

25%

Pilots & Operations

Qatar Free Zone or QFC entity incorporation, pilot deployment costs, and business development across energy and logistics.

A Complete “Full-Stack” Physical AI Leadership



Mohsen Mostafa

Co-Founder & Chief AI Scientist

The “Brain”: AI, Math, and MLOps.

Theorist: 8 first-author papers on geometric deep learning, Bayesian uncertainty (R-LayerNorm), and world models.

Engineer: 8+ years shipping production AI. Managed 24-GPU clusters, real-time Kafka/Spark pipelines, and TensorRT optimization.

Leader: Former AI Team Lead. Proven ability to architect enterprise AI systems from PoC to production.



Kaiz Ali

Co-Founder & Chief Hardware Engineer

The “Body”: Electronics, Sensors, and Physical Integration.

Expertise: HNC & HND qualified Electrical and Electronics Engineer.

Role: Sensor integration, edge compute hardware architecture, and ensuring Aventro’s AI models translate reliably into physical robotic actions.

Advantage: Ensures our software robotics solutions are grounded in electrical reality — capable of surviving harsh industrial and logistics environments.

STRATEGIC FIT

Why Qatar?

Aligning Aventro with Qatar National Vision 2030

1

Diversified Knowledge Economy

Our Physical AI platform directly serves Qatar's shift from a resource-based to a knowledge-driven economy — we produce foundational Physical AI research and technology from Qatar.

3

Hamad Port & Logistics Hub

Qatar has invested \$7B+ in Hamad Port, targeting top-10 global port status. Warehouse automation and real-time cargo anomaly detection are core to this ambition.

5

The UK-Qatar Corridor

As a British/Egyptian team based in the UK, we're the international deep-tech talent Qatar Free Zones and the QFC are designed to attract — backed by a strong, growing trade relationship.

2

Energy Sector Automation

QatarEnergy operates one of the world's largest LNG networks. Remote inspection and autonomous hazardous-zone robotics (ATEX-certified) are an immediate, high-value use case — unmatched GCC concentration of demand.

4

Post-World Cup Smart Infrastructure

Qatar built world-class smart city infrastructure for the 2022 FIFA World Cup. Maintaining these assets (stadia, metro, Lusail City) needs exactly the physical AI we build.

6

QCRI, HBKU & Qatar University

Our technical roadmap in geometric AI, world models, and uncertainty quantification makes us a natural collaborator for Qatar's world-class research institutions.

What We Will Achieve During the Startup Qatar Program



Giving Back to Qatar's Startup Ecosystem

1

Knowledge Sharing

Conducting workshops on “Resource-Efficient AI” for the Startup Qatar community and QCRI/HBKU researchers.

3

Talent Pipeline

Creating high-value, deep-tech AI jobs for Qatari nationals and expats, retaining talent that might otherwise move to the US, UK, or UAE. Active partnership with local universities for internships.

5

Regional Hub

Using Qatar's strategic geographic position and world-class logistics infrastructure as a base to serve the wider GCC and MENA market from Doha.

2

Open Source

Open-sourcing core uncertainty-aware layers (R-LayerNorm) and SafePredict modules — putting Qatar at the center of the Physical AI open-source community, not just consuming foreign AI tools.

4

Reputation

Establishing Qatar as the birthplace of Epistemology-Driven Robotics — a genuinely novel AI paradigm, not another LLM wrapper or generic CV startup.

Let's Build the Future of Physical AI Together

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APPENDIX AVAILABLE

Technical architecture diagrams · Math proofs for R-LayerNorm · Detailed MLOps pipeline schematics · Links to all 8 preprints